QUIZFORGE ACADEMIC ASSESSMENT

THERMODYNAMICS & HEAT TRANSFER - PRACTICE SET

Section I: Laws of Thermodynamics & Heat Engines

1. A reversible Carnot engine operates between temperatures of 800 K and 400 K. What is the theoretical maximum thermal efficiency of this engine?
(A) 25%
(B) 50%
(C) 66.7%
(D) 75%
Ans: (B)
Explanation: Thermal efficiency eta = 1 - (T_cold / T_hot) = 1 - (400/800) = 0.50 or 50%.

2. In an isothermal expansion of an ideal gas, which of the following thermodynamic state variables remains strictly constant?
(A) Enthalpy
(B) Pressure
(C) Temperature
(D) Entropy
Ans: (C)
Explanation: By definition, an isothermal process is one where the temperature remains constant (Delta T = 0).

3. According to the Second Law of Thermodynamics, the total entropy of an isolated system:
(A) Always decreases
(B) Must remain constant
(C) Can never decrease over time
(D) Oscillates around zero
Ans: (C)
Explanation: For any spontaneous process in an isolated system, the change in entropy Delta S >= 0.

4. A heat engine receives 1200 kJ of heat from a high-temperature reservoir and rejects 720 kJ to a low-temperature sink. What is the net work output of the engine in kJ?
(A) 480 kJ
(B) 600 kJ
(C) 720 kJ
(D) 1920 kJ
Ans: (A)
Explanation: W_net = Q_in - Q_out = 1200 kJ - 720 kJ = 480 kJ.

5. Consider the steady-state heat conduction across a plane wall of thickness L = 0.2 m with thermal conductivity k = 1.5 W/m-K and temperature difference Delta T = 40 K. Refer to the temperature profile diagram.
(A) 150 W/m^2
(B) 300 W/m^2
(C) 450 W/m^2
(D) 600 W/m^2
Ans: (B)
Explanation: Heat flux q = -k * (dT/dx) = 1.5 * (40 / 0.2) = 300 W/m^2.

ANSWER KEY & EXPLANATIONS

1. B
2. C
3. C
4. A
5. B